

Parkinson's Disease Drug Repurposing — Deep Validation (Retained Candidates)

CANDIDATE 1 — OLTIPRAZ

1. Candidate identity

Oltipraz — CHEBI:77319 (confirmed via Monarch); CAS 64224-21-1; synonym RP-35,972; a synthetic 1,2-dithiole-3-thione. Historically an FDA-investigated chemopreventive/antisteatotic agent (liver disease, lung cancer prevention). No history of PD-directed development.

2. Proposed mechanistic hypothesis

Oltipraz activates Nrf2 (NFE2L2), which induces NQO1/HO-1 expression and the canonical anti-ferroptotic module (GPX4 $\uparrow$, SLC7A11 $\uparrow$, ACSL4 $\downarrow$) in CNS tissue, reducing oxidative stress/ferroptosis and suppressing microglial activation. This CNS pharmacology is oltipraz-specific but was demonstrated only in acute-injury paradigms (a nerve-injury/neuropathic-pain model and a cerebral ischemia-reperfusion model) — never in a PD, MPTP, 6-OHDA, or α -synuclein model. The link to PD is an inferred bridge: the same Nrf2 $\rightarrow$ GPX4/NQO1 axis and anti-microglial effect are independently protective for dopaminergic neurons in actual PD models, but there driven by different compounds (melatonin, schisanhenol, β -caryophyllene).

This revises the originally-proposed initial mechanism. The initial framing — oltipraz acting through a "parallel non-mitochondrial FSP1–CoQ10 axis" — has **no oltipraz-specific evidentiary support** in PubMed at all. The real, verifiable oltipraz-CNS pharmacology is

GPX4/SLC7A11/ACSL4-centric (the canonical ferroptosis-defense axis), not FSP1/CoQ10-centric. This is a genuine mechanism revision, not a refinement.

3. Complete mechanism chain

1. Oltipraz → activates Nrf2 (NFE2L2)
2. Nrf2 → induces NQO1/HO-1 in CNS tissue (spinal cord, hippocampus, prefrontal cortex)
3. Nrf2 → upregulates GPX4/SLC7A11, downregulates ACSL4 (antiferroptotic program), in brain tissue (MCAO/R model) and PC12 cells
4. Oltipraz (via this axis) → suppresses microglial activation (CD11b/c↓) in CNS tissue
5. This Nrf2-driven axis → reduces oxidative stress / ferroptosis in neural tissue (oltipraz-specific, non-PD models)
6. **[Inference]** The same Nrf2→GPX4/NQO1 axis and anti-microglial effect, independently shown (other compounds) to protect dopaminergic neurons in PD/MPTP/α-synuclein models → hypothesized to translate to oltipraz's own PD-relevant neuroprotection — **never directly tested**

Secondary, weaker corroborating strand (target-class, not oltipraz-specific): NQO1 interacts with PRKN (Parkin, a bona fide causal PD gene per Monarch's curated OMIM/Orphanet associations) and AKT1; Akt-mediated phosphorylation of NQO1 triggers Parkin-mediated degradation, reducing NQO1's antioxidant capacity as PD progresses.

4. Edge-Level Evidence Ledger

#	Subject	Predicate	Object	Evidence status	Source	Identifier	Location	Evidence statement
1	Oltipraz	activates	Nrf2 (NFE2L2)	OBSERVED_LITERATURE	PubMed	PMID 37535075	ABSTRACT	Oltipraz described as an Nrf2 activator inducing NQO1/GCL, protecting mouse liver

#	Subject	Predicate	Object	Evidence status	Source	Identifier	Location	Evidence statement
								from acetaminophen injury via GSH recovery (non-CNS confirmation of core pharmacology).
2	Oltipraz	induces (Nrf2-dependent), CNS tissue	NQO1, HO-1 (spinal cord, hippocampus, PFC)	OBSERVED_LITERATURE	PubMed	PMID 31234342	ABSTRACT	Mouse sciatic-nerve-injury (neuropathic pain) model: systemic oltipraz enhanced Nrf2/HO-1/NQO1 in spinal cord, hippocampus, PFC — confirm

#	Subject	Predicate	Object	Evidence status	Source	Identifier	Location	Evidence statement
								s oltipraz reaches and engages this pathway in CNS tissue. Also decreased p-Akt/PI3K signaling. Not a PD model.
3	Oltipraz	upregulates GPX4/SLC7A11, downregulates ACSL4, Nrf2-dependently	antiferroptotic program (brain, MCAO/R; PC12 cells)	OBSERVED_LITERATURE	PubMed	PMID 39096875	ABSTRACT	Oltipraz reduced infarct volume/oxidative stress/ferroptosis in mouse cerebral ischemia-reperfusion; Nrf2-KO abolished the effect

#	Subject	Predicate	Object	Evidence status	Source	Identifier	Location	Evidence statement
								(causal) · Canonical GPX4-centric axis, not FSP1/CoQ10. Not a PD model.
4	Oltipraz	suppresses	microglial activation (CD11b/c), CNS tissue	OBSERVED_LITERATURE	PubMed	PMID 31234342	ABSTRACT	Same neuropathic-pain study: oltipraz inhibited microglial activation and PI3K/Akt, NF-κB, MAPK overexpression in spinal cord. Not a PD model.
5	GPX4-centric antiferro	protects	dopaminergic neurons	OBSERVED_LITERATURE	PubMed	PMID 38488331	ABSTRACT	Different compound

#	Subject	Predicate	Object	Evidence status	Source	Identifier	Location	Evidence statement
	ptotic activity (Nrf2-driven, other compounds)		, PD/MPTP models	URE (class-level, NOT oltipraz)		(melatonin, Sirt1/Nrf2/HO-1/GPX4, α -syn PFF model — NQO1 not tested); PMID 41460594 (schishenol, Nrf2/TrxR1/GPX4, MPTP model — SLC7A11 not involved)		nds; NQO1/SLC7A11 combination oltipraz engages is not reproduced identically in either bridge paper — "same axis" framing overstates mechanistic overlap (critical finding).
6	NQO1 activity (other compounds / endogenous)	supports	dopaminergic neuron survival, PD models	OBSERVED_LITERATURE (class-level, NOT oltipraz)	PubMed	PMID 33271195 (β -caryophyllene, MPTP model); PMID 313586	ABSTRACT	Neither paper involves oltipraz. PMID 31358653's stage-dependent

#	Subject	Predicate	Object	Evidence status	Source	Identifier	Location	Evidence statement
						53 (NQO1 neuroprotective in MPTP + α -syn-transgenic mice; Akt phosphorylation $\rightarrow$ Parkin-mediated degradation abolishes protection; NQO1 protein "most prominent in active/early-stage PD, virtually absent at end-stage")		nt NQO1 finding raises a therapeutic-window concern independent of any delivery question.

#	Subject	Predicate	Object	Evidence status	Source	Identifier	Location	Evidence statement
7	NQO1 (HGNC: 2874)	interacts_with	PRKN (HGNC: 8607), AKT1	OBSERVED_KG — but not independent (see Critic finding)	Monarch	HGNC: 2874 ↔ HGNC: 8607 / AKT1, infores: biogrid	MONARCH_KG	BioGRID interaction edges. On inspection, the underlying supporting publications are PMID 31358653 (already cited directly) plus two PD-irrelevant cancer-biology papers (PMID 26540344 breast cancer/p53; PMID 31842909 hepatocellular

#	Subject	Predicate	Object	Evidence status	Source	Identifier	Location	Evidence statement
								carcinoma/SIRT6) — this KG strand is largely circular re-indexing of the same primary evidence, not a genuinely independent corroboration.
8	PRKN (HGNC: 8607)	causes / gene_associated_with	Autosomal recessive juvenile PD 2 (MONDO:0010820); Young-onset PD (MONDO:0017279)	OBSERVED_KG	Monarch	HGNC: 8607↔ MONDO:0010820 (OMIM, causal), ↔MONDO:0017279 (Orphanet)	MONARCH_KG	Confirms PRKN is a genuinely well-annotated, bona fide causal PD gene (36 phenotype

#	Subject	Predicate	Object	Evidence status	Source	Identifier	Location	Evidence statement
								associations including parkinsonism, Lewy bodies, SN gliosis, bradykinesia).
9	Oltipraz	(no relationship found in literature to)	PD / dopaminergic neurons / MPTP / 6-OHDA / SN / α -synuclein	UNRESOLVED (confirmed absence, twice-independently reproduced)	PubMed	zero-hit search, reconfirmed by both the reconstruction agent and the critic	ABSTRACT (search listing)	No compound-specific PD evidence exists anywhere in PubMed for oltipraz.
10	NQO1 C609T polymorphism (rs1800566)	NOT significantly associated with	Parkinson's disease	CONTRADICTED (corrects a prior mischaracterization)	PubMed	PMID 25054016	ABSTRACT	Meta-analysis, 15 studies (2,858 cases/2,907 controls): null result for NQO1; only

#	Subject	Predicate	Object	Evidence status	Source	Identifier	Location	Evidence statement
								TNF-1031 was significant. Prior screening cited this PMID as "NQO1 genetic association meta-analysis" supporting the mechanism — that characterization is incorrect; the actual finding does not support an NQO1-PD genetic link.

#	Subject	Predicate	Object	Evidence status	Source	Identifier	Location	Evidence statement
11	[NEW — human PD-relevant target engagement] Oral oltipraz (200–400 mg/wk, 12 weeks)	failed to engage	Nrf2/phase-II pathway in human tissue (chronic smokers, lung epithelium)	OBSERVED_LITERATURE, CONTRADICTED	PubMed	PMID 15824161	ABSTRACT	"No increase in mRNA or enzymatic activity of phase II enzymes and no change in glutathione levels," despite 15% grade 2/3 (mostly GI) toxicity and all study withdrawals occurring in the oltipraz arms. Peripheral tissue, not

#	Subject	Predicate	Object	Evidence status	Source	Identifier	Location	Evidence statement
								brain — but directly undercuts the assumption that oral dosing reliably drives oltipraz' s own proposed target engagement in humans at all. Surface d independently by the Critic, not the original reconstruction.
12	Oltipraz	has documented off-target activity via	LXR- α inhibition (antistatotic mechanism in	OBSERVED_LITERATURE, unaddressed	PubMed / Clinical Trials.gov	PMID 28225186 / NCT01373554	ABSTRACT	LXR- α is expressed in microglia and regulate

#	Subject	Predicate	Object	Evidence status	Source	Identifier	Location	Evidence statement
			NAFLD, same 30–120 mg/day dose range)	confounded				s CNS lipid/inflammatory tone — an unacknowledged confound for asserting the CNS effect is Nrf2-specific. Not evidence of harm, but a real gap in the mechanism's precision.

5. Agent-inferred bridges and their reasoning

Central bridge (Edge 6): Oltipraz's own confirmed CNS pharmacology (Edges 1–5: Nrf2 activation → NQO1/HO-1 + GPX4/SLC7A11/ACSL4 anti-ferroptotic program + anti-microglial effect, demonstrated in vivo in two independent CNS injury models) is bridged to PD-relevant dopaminergic neuroprotection (the same axis, shown protective for dopaminergic neurons in actual PD/MPTP models — but there driven by melatonin, schisanhenol, or β -caryophyllene, never oltipraz). Justification: the convergence of two independent oltipraz-specific studies (different investigators, different injury models) on the same Nrf2-driven signature suggests this is oltipraz's real, reproducible CNS pharmacology, and that same signature is independently

protective in dopaminergic-neuron contexts. Why it remains inference: no study has ever dosed oltipraz in any PD-relevant model; the CNS evidence is from acute nerve-injury and stroke paradigms in non-nigral tissue, not chronic degeneration in substantia nigra dopaminergic neurons — a real biological gap, not just an absence of data (Critic finding).

Secondary, weaker bridge (Edge 7): treated by the reconstruction agent as corroborating but shown by the Critic to be largely circular (re-indexed citations of the same primary paper plus PD-irrelevant cancer literature) — this strand should carry little independent evidentiary weight.

6. Unresolved relationships

- ClinicalTrials.gov independent re-verification of the "6 trials, all liver/lung, zero PD" finding: **SOURCE_ACCESS_FAILED** in the reconstruction pass (official [/api/v2/](#) blocked by robots.txt + org egress policy); the Critic independently reconfirmed the trial list via the same underlying data referenced in the N/P critic pass, but could not run a fresh independent CT.gov query this session either.
- Whether oltipraz achieves meaningful CNS/brain exposure at doses relevant to any of its human trials (all peripheral-indication trials; no CNS PK study located in the approved sources) — **UNRESOLVED**.
- Whether Nrf2 or NQO1 carry any curated Monarch gene-disease association with PD — **resolved as absent** (confirmed by direct query, not a search failure); this doesn't contradict the literature-based mechanism but means it cannot currently be corroborated through Monarch's curated layer.

7. Contradictory evidence

- PMID 25054016 does not support an NQO1-PD genetic association (null result) — corrects a screening mischaracterization (see Edge 10).
- The initial "FSP1–CoQ10 axis" framing has zero oltipraz-specific support — the real evidence is GPX4/SLC7A11/ACSL4-centric (Edge 3), a genuine mechanism revision.
- PMIDs 33271195 and 38455026, previously cited at screening as if supporting oltipraz, are actually about different compounds (β -caryophyllene, inflachromene) — target-class evidence only.
- **New (from Critic):** PMID 15824161 shows oral oltipraz failed to pharmacodynamically engage its own proposed Nrf2/phase-II target in a human trial, with a meaningful toxicity/withdrawal signal — this is real evidence against reliable human target engagement, independent of the CNS-penetration question.
- **New (from Critic):** the Monarch KG "corroborating strand" (NQO1-PRKN/AKT1 interactions) is largely circular, not independent evidence.
- No direct contradiction was found against oltipraz's core CNS pharmacology itself (Nrf2 activation, NQO1/HO-1/GPX4 induction, anti-microglial effect) — this compound-specific

evidence held up under independent re-verification by both the reconstruction agent and the Critic.

8. Independent Critic findings

All core PMIDs verified as real and accurately characterized; the reconstruction agent's self-corrections (dropping FSP1/CoQ10, nulling PMID 25054016, relabeling target-class-only citations) were accurate and appropriately conservative. The Critic's own independent contribution surfaced four new weaknesses not in the original reconstruction: (1) the oltipraz-specific CNS evidence comes only from acute-injury paradigms in non-nigral tissue — a real translational gap for a chronic, region-selective disease; (2) oral oltipraz failed to engage its own proposed target pharmacodynamically in the one human trial that measured it (PMID 15824161), with a real toxicity signal; (3) the Monarch KG "corroborating strand" is largely circular rather than independent; (4) NQO1's PD-stage-dependent expression (high early, absent late per PMID 31358653) raises a therapeutic-window concern. The Critic's overall read: the target-class hypothesis (Nrf2/antioxidant induction in PD) is legitimate, but the oltipraz-specific case is thinner than the reconstruction's framing conveyed.

9. Main remaining evidence gap

No study has ever tested oltipraz in any PD-relevant model (cellular, animal, or human), and the one human trial that measured whether oral oltipraz reliably engages its own proposed Nrf2 target found that it does not — at doses/routes not designed for or tested against CNS exposure. Whether oltipraz can achieve CNS target engagement at a tolerable dose, in a form relevant to chronic dopaminergic-neuron protection rather than acute injury response, is entirely open.

10. Final verdict

INCONCLUSIVE

11. Justification

Oltipraz's own CNS-relevant pharmacology (Nrf2 activation, NQO1/HO-1 induction, GPX4/SLC7A11/ACSL4 anti-ferroptotic engagement, anti-microglial suppression) is real, oltipraz-specific, and reproduced across two independent injury models — this is solid ground, not speculation. The link to PD specifically, however, rests entirely on an untested cross-model, cross-tissue extrapolation (acute nerve-injury/stroke pharmacology in non-nigral tissue → chronic dopaminergic-neuron protection), reinforced by a target-class literature (other compounds, same pathway, real PD models) rather than any oltipraz-specific PD data. This alone would support "plausible but unproven." What pushes it to INCONCLUSIVE rather than a stronger HOLDS_UP reading is a genuine negative signal the Critic surfaced: the one human

trial measuring oltipraz's own target engagement found it failed to occur, alongside a real toxicity/withdrawal signal — a concrete reason to doubt reliable target engagement independent of the (still entirely untested) blood-brain-barrier question. Nothing found actively falsifies the hypothesis (no evidence oltipraz harms dopaminergic neurons or opposes the proposed direction), so HYPOTHESIS_UNDERMINED is not warranted — but the evidence stops well short of confirming a viable path to PD-relevant efficacy.

CANDIDATE 2 — PMX205

1. Candidate identity

PMX205 — cyclic hexapeptide, small-molecule antagonist of C5aR1 (CD88, complement C5a receptor 1), structurally related to PMX53. No brand name; still preclinical (zero registered clinical trials, independently reconfirmed twice this session via ClinicalTrials.gov's internal search API, since the public [/api/v2/](#) and [/search?](#) endpoints are robots-disallowed / egress-blocked on this network).

2. Proposed mechanistic hypothesis

α -Synuclein aggregation activates the classical complement pathway (C1q-initiated), proceeding through C3/C5 cleavage to cause complement-dependent cytotoxicity and, separately, microglial synaptic engulfment of dopaminergic circuitry in PD models. PMX205, a validated brain-active C5aR1 antagonist, is hypothesized to interrupt the C5a/C5aR1 arm of this signaling, dampening microglial/astrocyte activation — based entirely on PMX205's validated efficacy in four *other* CNS neuroinflammatory disease models (Alzheimer's, hypoxic-ischemic injury, intracerebral hemorrhage, radiation-induced cognitive decline), not on any PD or α -synuclein data point.

This mechanism is more precisely characterized, and more seriously undermined, than the original screening framing conveyed. Two independent investigations (reconstruction + Critic) converge on the same critical structural problem: the one PD-model paper that directly links complement to synaptic pruning implicates **C3-C3aR**, not C5-C5aR1 — PMX205 does not target the receptor associated with PD-model synaptic engulfment. The PD genetic model with the clearest synaptic-pruning phenotype (LRRK2 G2019S) implicates no complement component at all. And the one study that directly manipulates a complement gene in an α -synuclein-relevant model found that **removing C1q worsened**, not improved, the pathology.

3. Complete mechanism chain

1. α -Synuclein aggregation → activates classical complement pathway (C1q-initiated) — shown in α -syn-expressing cells, with C1q trend elevated in postmortem **MSA** (not PD) putamen
2. Classical pathway activation → C3/C5 cleavage → complement-dependent cytotoxicity (rescued in vitro by C3- and C5-specific inhibitors — but this data cannot distinguish the C5a/C5aR1 arm from the C5b-9 membrane-attack-complex arm)
3. C5 cleavage → generates C5a (canonical biochemistry, C5a is the C5aR1 ligand)
4. C5a–C5aR1 signaling → activates microglia/astrocytes (general CNS biology, GO-confirmed for C5AR1; zero curated PD-specific association)
5. Microglial/complement-linked synaptic engulfment of dopaminergic circuitry → shown in a **C3-C3aR-driven** rotenone PD toxin model (different receptor than PMX205's target) and, separately (no complement link at all), in a LRRK2 G2019S genetic PD model
6. PMX205 → antagonizes C5aR1 (well-established pharmacology, multiple independent labs)
7. PMX205/C5aR1 antagonism → suppresses microglial/astrocyte activation in four other CNS disease models (none is PD; none examines nigral/nigrostriatal circuitry)
8. **[Inference — the central, unresolved leap]** PMX205's C5aR1 blockade → hypothesized to dampen complement-driven dopaminergic-circuit engulfment in PD — **never tested; targets a different receptor than the one PD-model paper that shows complement-linked synaptic engulfment**

4. Edge-Level Evidence Ledger

#	Subject	Predicate	Object	Evidence status	Source	Identifier	Location	Evidence statement
1	α -Synuclein	activates	classical complement pathway (C1q)	OBSERVED_LITERATURE	PubMed	PMID 34399786	ABSTRACT	α -syn-expressing SH-SY5Y cells; C1q trend elevated specifically in MSA

#	Subject	Predicate	Object	Evidence status	Source	Identifier	Location	Evidence statement
								postmortem putamen (not PD tissue). Paper frames PD/DLB/MSA together as synucleinopathies — treating the MSA data as strong PD-specific human evidence is a bigger inferential leap than the "synucleinopathy" framing suggests (MSA α -syn is predomi

#	Subject	Predicate	Object	Evidence status	Source	Identifier	Location	Evidence statement
								namely oligodendrocytic, pathologically distinct from PD's neuronal Lewy pathology).
2	Complement activation (C3/C5)	mediate s	cytotoxicity in α -syn-expressing cells	OBSERVED_LITERATURE	PubMed	PMID 34399786	ABSTRACT	Rescued by C5-inhibitor RaCl and C3-inhibitor Cp20 — C3/C5-dependent, not merely correlative. In vitro cell-line model only. RaCl blocks C5 cleavage

#	Subject	Predicate	Object	Evidence status	Source	Identifier	Location	Evidence statement
								e itself, not the C5aR1 receptor — cannot isolate the C5a/C5aR1 arm from the C5b-9 (membrane-attack-complex) arm.
3	C5 cleavage	generates	C5a (C5aR1 ligand)	SUPPORTED_CONTEXT	Monarch	HGNC: 1338 (C5AR1), GO:0038178 / GO:0004878	MONARCH_KG	Canonical complement biochemistry (GO-annotated), general, not PD-specific.
4	C5aR1	activates	microglia / astrocytes	OBSERVED_KG	Monarch	HGNC: 1338, GO:0001774 / GO:004	MONARCH_KG	General biological-process annotated

#	Subject	Predicate	Object	Evidence status	Source	Identifier	Location	Evidence statement
						8143 / GO:0010759		on. Zero disease association attached — but this is a curation-coverage artifact, not a PD-specific signal: C5AR1 has zero curated gene-disease associations in Monarch for <i>any</i> disease, including Alzheimer's, despite ≥8 independent PubMed papers

#	Subject	Predicate	Object	Evidence status	Source	Identifier	Location	Evidence statement
								linking PMX205/C5aR1 to AD models.
5a	Microglia	prune/engulf	dopaminergic fibers (LRRK2 G2019S PD genetic model)	OBSERVED_LITERATURE	PubMed	PMID 34409578	ABSTRACT (directly re-read by Critic)	Confirmed: abnormal microglial phagocytosis/pruning of DAergic fibers, inflammatory transcriptional remodeling, PD-related behavioral deficits. Zero mention of C1q/C3/C5/C5aR1 anywhere in this

#	Subject	Predicate	Object	Evidence status	Source	Identifier	Location	Evidence statement
								paper — the clearest PD-model synaptic-pruning evidence available does not implicate complement at all.
5b	C3-C3aR axis	drives	synaptic engulfment / dark microglia (rotenone PD toxin model)	OBSERVED_LITERATURE (title only)	PubMed	PMID 41637879	ABSTRACT title (full text SOURCE_ACCESS_FAILED after repeated retries by both reconstruction and Critic)	Title confirms a complement-linked PD-model synaptic-engulfment finding — but implicates C3 and C3aR , a different

#	Subject	Predicate	Object	Evidence status	Source	Identifier	Location	Evidence statement
								receptor system than C5/C5a R1. PMX205 does not target this receptor.
6	PMX205	antagonizes	C5aR1	OBSERVED_LITERATURE	PubMed	PMID 39147742, 41967875, 42570158, 32064396(†)	ABSTRACT (titles independently confirmed)	Well-established, reproducible pharmacology across independent labs. (†) PMID 32064396 (the PK study) could not be relocated by the Critic across 6

#	Subject	Predicate	Object	Evidence status	Source	Identifier	Location	Evidence statement
								independent topical searches — flagged as a corroboration gap, not a fabrication.
7a	PMX205 (C5aR1 antagonism)	suppresses	inflammatory glial responses — Alzheimer's model	OBSERVED_LITERATURE	PubMed	PMID 39147742 (+ preprint duplicate 37662399)	ABSTRACT title	AD model, not PD. Critic independently surfaced ≥6 additional PMX205/C5aR1-AD-adjacent papers not in the original citation list — the pharmacological

#	Subject	Predicate	Object	Evidence status	Source	Identifier	Location	Evidence statement
								I tool is on very solid ground generally.
7b	PMX205 (C5aR1 antagonist)	confers neuroprotection in	hypoxic-ischemic brain injury	OBSERVED_LITERATURE	PubMed	PMID 41967875	ABSTRACT title	Not PD; different brain regions/injury type than nigral degeneration.
7c	PMX205 (C5aR1 antagonist)	attenuates	neuroinflammation — intracerebral hemorrhage	OBSERVED_LITERATURE	PubMed	PMID 42570158	ABSTRACT title	Not PD; perihematomal region.
7d	C5aR1 inhibition	alleviates	cranial radiation-induced cognitive decline	OBSERVED_LITERATURE	PubMed	PMID 41085542 (peer-reviewed) = PMID 39005286 (bioRxiv preprint) — confirm	ABSTRACT title	Not PD; hippocampal/white-matter region. The "four independent CNS-disease

#	Subject	Predicate	Object	Evidence status	Source	Identifier	Location	Evidence statement
						ed same study, not two independent findings		models" is accurate as four <i>studies</i> , one (radiation) reported twice across preprint → publication.
8	PMX205	tested in	PD / α -synuclein model	CONTRADICTED (absence, independently reproduced twice)	PubMed	"PMX205 Parkinson," "PMX205 alpha-synuclein," "C5aR1 Parkinson" — all zero/near-zero on-topic hits	ABSTRACT (search listing)	Given how populated adjacent literatures are (PD-C1q, PD-C3aR, AD-C5aR1 each well-represented independently), this specific

#	Subject	Predicate	Object	Evidence status	Source	Identifier	Location	Evidence statement
								intersection being empty reads as a genuine , searchable research gap, not mere obscurity (Critical finding).
9	C5AR1	associated with	Parkinson disease	CONTRADICTED (absence of KG support — but weak evidence, see Edge 4)	Monarch	HGNC: 1338 vs MONDO:0005180, total: 0	MONARCH_KG	PD's curated causal/correlated gene list in Monarch (SNCA, PRKN, PINK1, PARK7, LRRK2, DNAJC6, ATP13A2, FBXO7, SYNJ1,

#	Subject	Predicate	Object	Evidence status	Source	Identifier	Location	Evidence statement
								PTPA, PLA2G6, CHCHD2, VPS13C, GBA1, MAPT, VPS35) contains no complement genes at all.
10	[NEW — direct contradicting evidence] Genetic C1q deletion	worsens	enteric α -synuclein neuropathology (while improving gut motility)	CONTRADICTED — directly complicates the core premise	PubMed	PMID 37961460 (preprint, bioRxiv)	ABSTRACT	The one study that directly manipulates a complement gene in an α -synuclein-relevant model found complement engulfment/exhaustion

#	Subject	Predicate	Object	Evidence status	Source	Identifier	Location	Evidence statement
								of ENS-resident macrophages was protective/clearance-oriented for neurons, not purely pathogenic — a real dissociation. Peripheral (enteric), not central, so generalizability to nigral circuits is uncertain, but it directly challenges the "complement

#	Subject	Predicate	Object	Evidence status	Source	Identifier	Location	Evidence statement
								activation = bad, block it = good" premise underlying the PMX205 hypothesis.

5. Agent-inferred bridges and their reasoning

Bridge A (Edge 3): general complement biochemistry (C5→C5a→C5aR1) applied to the α-synuclein/complement-cytotoxicity finding (Edge 2) — reasonable canonical biochemistry, but PMID 34399786's C5-inhibition data cannot itself distinguish the C5a/C5aR1 signaling arm from the C5b-9 membrane-attack-complex arm, so this is inference from general pathway structure, not a demonstrated finding.

Bridge B (Edge 8, the central and unavoidable leap): bridges (i) real synucleinopathy-model evidence that α-syn triggers complement-dependent cytotoxicity (Edges 1–2, MSA tissue + cell-line data), (ii) KG-confirmed general biology that C5aR1 activates microglia (Edge 4), (iii) PD-model evidence of complement-linked synaptic engulfment — but via a *different receptor*, C3aR, not C5aR1 (Edge 5b), and separately, complement-silent synaptic pruning in a genetic PD model (Edge 5a), and (iv) PMX205's validated efficacy suppressing C5aR1-driven neuroinflammation in four non-PD CNS diseases (Edges 6–7d). No single study connects these; no PD model, no α-synuclein model, and no C5aR1-specific tool (genetic or PMX205) have ever been combined. Justified as a plausible, testable hypothesis by the convergence of independently real findings — but the receptor-level mismatch between PMX205's actual target and the one complement receptor shown relevant to PD-model synaptic pruning is a specific, previously underweighted structural weakness in this bridge, not just a general absence of data.

6. Unresolved relationships

- PMID 32064396 (PK study, cited as evidence of CNS-penetrant peripheral dosing) — could not be relocated by the Critic in 6 independent topical searches; **UNRESOLVED**, not fabricated, but not independently corroborated either.

- PMID 41637879 full text (the C3-C3aR rotenone PD-model paper) — **SOURCE_ACCESS_FAILED** after repeated retries by both agents; only the title is confirmed.
- ClinicalTrials.gov: public [/api/v2/](#) and [/search?](#) endpoints blocked (robots.txt + org egress policy); the permitted [/api/int/](#) endpoint was used instead and returned an authoritative **total: 0** for PMX205, independently reconfirmed — treated as resolved via an alternate access *method* for the same approved source, not a substitution.
- Google Patents' independent confirmation of PMX205's patent-novelty status remains outside this Deep Validation's approved-source scope (PubMed/Monarch/CT.gov only) and stays an open item from the earlier N/P critic.

7. Contradictory evidence

- The clearest PD-model synaptic-pruning paper (LRRK2 G2019S, PMID 34409578) does not implicate complement at all — weakens, without falsifying, the premise that PD-relevant synaptic pruning is complement-driven.
- The one PD-model paper that does link complement to synaptic engulfment implicates **C3-C3aR**, a different receptor than PMX205's target, C5aR1 (PMID 41637879, title-level).
- **PMID 37961460: genetic C1q deletion worsened enteric α -synuclein pathology** — direct evidence complicating the simple "complement=bad" framing in the one study that directly manipulates a complement gene in an α -synuclein-relevant model.
- Monarch shows zero curated PD-C5aR1 association — but this is shown to be a weak signal on inspection, since the same zero-association result holds for Alzheimer's disease despite abundant AD-C5aR1 literature (curation-coverage artifact, not a PD-specific finding).
- All originally-cited PMIDs were confirmed real and accurately titled; no fabricated or misrepresented citations were found by either agent.

8. Independent Critic findings

The Critic independently verified nearly every cited PMID and found PMX205's core pharmacology (C5aR1 antagonism, multi-model CNS anti-inflammatory efficacy) even better-supported than originally cited, surfacing ≥ 8 additional corroborating papers. But the Critic's own contribution was to sharpen the central weakness: the receptor-level mismatch between PMX205 (C5aR1) and the one PD-model complement-pruning paper (C3aR) is a structural problem, not just "PMX205 untested in PD" — the one complement receptor with any PD-model synaptic-pruning association is not the one PMX205 targets. The Critic also found that none of PMX205's four supporting CNS-disease efficacy models examine the substantia nigra / nigrostriatal circuitry central to PD, and that the Monarch "zero PD association" is a weaker signal than presented (same zero result for AD, a curation-coverage issue). The Critic's overall read: a defensible, testable hypothesis, but resting on a longer and shakier inferential chain — spanning a cross-disease (MSA $\rightarrow$ PD) tissue extrapolation, a receptor-mismatch in the

one PD-relevant complement finding, and cross-region (non-nigral) efficacy data — than the reconstruction's confident framing conveyed.

9. Main remaining evidence gap

No study has combined PMX205, or any C5aR1-specific genetic or pharmacological tool, with a PD or α -synuclein model. The one complement receptor independently linked to PD-model synaptic engulfment (C3aR) is not the receptor PMX205 blocks (C5aR1) — whether C5aR1 blockade would meaningfully affect PD-relevant dopaminergic-circuit pathology, given this receptor-specific gap, is entirely untested.

10. Final verdict

HYPOTHESIS_UNDERMINED

11. Justification

PMX205 itself is a well-validated, CNS-active C5aR1 antagonist with genuinely broad and reproducible anti-neuroinflammatory efficacy — that part of the case is solid. But the specific mechanistic bridge to PD is undermined by more than an absence of data: the one PD-model study that directly links complement activation to synaptic engulfment implicates a *different* complement receptor (C3aR) than the one PMX205 targets (C5aR1); the PD-model paper with the clearest synaptic-pruning phenotype (LRRK2 G2019S) implicates no complement component at all; and the one study that directly manipulates a complement gene in an α -synuclein-relevant context found that removing complement (C1q) *worsened* rather than improved the pathology — directly complicating the "block complement, protect neurons" premise the hypothesis depends on. None of PMX205's four supporting efficacy models examines the nigral/nigrostriatal circuitry central to PD, and the human-tissue evidence for α -synuclein-complement activation (MSA putamen) comes from a related but pathologically distinct synucleinopathy, not PD itself. Taken together — receptor-level mismatch, complement-silent PD-model pruning evidence, and a directly contradicting genetic finding — this is not simply an untested-but-plausible extrapolation; specific, real evidence weighs against the proposed mechanistic bridge as currently framed, even though PMX205's own pharmacology and the general complement-neuroinflammation-PD research area remain legitimate.

Explicit note on conflicts with prior screening (per instruction: do not silently overwrite)

- **Oltipraz:** initial screening's M3/F2/Direction=consistent/CNS=feasible ratings, and its citation of PMID 25054016 as NQO1 genetic-association support, are **not fully supported** by this Deep Validation pass. The FSP1-CoQ10 mechanism framing is revised (no oltipraz-specific support found); PMID 25054016 is a null result, not positive evidence; and new evidence (PMID 15824161) shows oral oltipraz failed human target engagement in the one trial that measured it. These are presented as revisions arising from independent verification, not as a rejection of the underlying retention decision — the critic-vetted N3/N4, P4 novelty status is unaffected and still stands.
- **PMX205:** initial screening's Direction=consistent and CNS=feasible ratings are **challenged** by this Deep Validation pass. New evidence not present in the initial screening — the C3aR-not-C5aR1 receptor mismatch in the one PD-model complement-pruning paper, and the C1q-deletion-worsens-pathology finding — was not available to or considered in the original screening and materially weakens the mechanistic case, independent of the (unaffected) N3/P4 novelty status.

STOP — Deep Validation complete for PMX205 and Oltipraz only. No further candidates investigated.